Software Requirements Specification (SRS) & Product Requirements Document (PRD)

Document Version: 1.0.0

Status: Approved for Architecture & Implementation

Target System: Internal Hybrid Project & Financial Management Platform

1. Executive Summary & Objective

1.1 Purpose

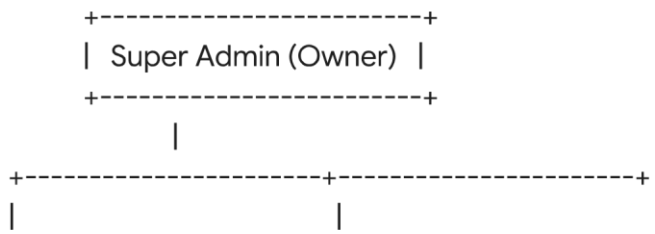
This document provides the foundational engineering and business specifications for an integrated, internal-first **Project and Financial Management System**. The platform resolves the operational overhead of managing blended teams—comprising full-time onsite employees and external percentage-partner freelancers—within a unified operational and financial ledger.

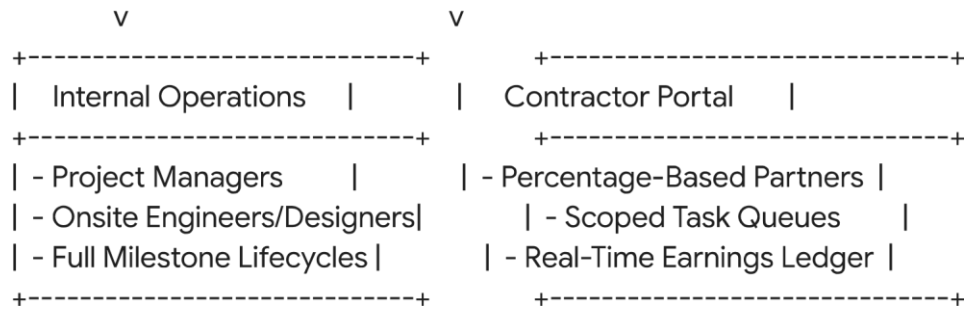
1.2 Core Business Model

- **Internal Team Operations:** Fixed-cost execution tracking, internal accountability, and task milestone velocity.
- **Freelancer Partnership Operations:** Commission- and milestone-based model where external contractors work on a pre-negotiated percentage of total project value or milestone billing.
- **Accountability & Governance:** Strict role-based access control (RBAC) paired with an immutable audit log engine tracking individual interactions, task progress, and item completions.

2. System Architecture & Role-Based Access Control (RBAC)

The application enforces strict separation of concerns via four primary operational roles across three core interfaces.





2.1 Role-Permission Matrix

Functional Capability	Super Admin (Owner)	Project Manager (Onsite)	Onsite Member	Partner
System User Provisioning	Full (CRUD)	None	None	None
Project Creation & Deletion	Full (CRUD)	Create & Edit Assigned	View Assigned	None
Freelancer Onboarding & % Setup	Full	View Only (Optional)	None	None
Task Management (Create/Edit/ Delete)	Full	Full (Assigned Projects)	None	None
Task Status & Checklist Completion	Full	Full	Assigned Only	Assigned Only
Audit Log Inspection	Global	Project Scoped	Personal Only	Personal Only

Client Invoicing & Total Contract Values	Full	Scoped / Read-Only	None	None

Freelancer Payout Calculation	Automated & Editable	Read-Only	None	Read-Only (Own)
Disbursement Authorization	Full (Sign-off)	None	None	None

3. Detailed Functional Modules

3.1 Super Admin Module (Ownership & Governance)

- **Contractor & Partner Setup:**
  - Define contractor profiles with legal identity, tax classification, and default payout channels.
  - Assign partnership models per project:
    - *Gross Contract Share:* Percentage calculated against total client invoice.
    - *Net Revenue Share:* Percentage calculated after deducting defined direct project expenses.
    - *Milestone Tranche Share:* Variable percentage tied to discrete milestone deliveries.
- **Global Financial Overview:**
  - Live monitoring of Total Contract Value (TCV), Gross Invoiced, Accounts Receivable, and Accrued Contractor Liabilities.
  - Payout release approval: Two-step authorization mechanism before marking any contractor balance as disbursed.

3.2 Project Management & Onsite Team Module

- **Project Structuring:**
  - Project hierarchy: Client Portfolio $\rightarrow$ Project $\rightarrow$ Milestones $\rightarrow$ Tasks $\rightarrow$ Checklist Items.
  - Target delivery dates, budgets, priorities (Urgent, High, Medium, Low), and resource allocation.
- **Task Lifecycle Engine:**
  - Kanban, List, and Gantt-style interfaces.
  - Dedicated actions for managers: Create, Update, Assign, Block, Archive, and Soft-Delete.
  - State transitions: Backlog $\rightarrow$ Assigned $\rightarrow$ In Progress $\rightarrow$ Pending Review $\rightarrow$ Completed.
- **Onsite Collaboration:**
  - Rich-text descriptions, file attachments, and internal team commenting threads.

3.3 Freelancer Partner Module

- **Data-Isolated Workspace:**
  - Access restricted strictly to projects and tasks explicitly assigned to the partner.
  - Strict concealment of client-facing commercial rates (unless identical to calculation basis), other team members' financial compensations, and internal team performance metrics.
- **Deliverable Pipeline:**
  - Ability to claim assigned checklist items, submit asset links or code commits, and transition tasks to Pending Review.
- **Partner Financial Ledger:**
  - Live dashboard reflecting earned vs. unbilled milestone commissions.
  - Transparent computation engine:

$$\text{Milestone Payout} = \text{Billed Milestone Amount} \times \left(\frac{\text{Contractor Agreement \%}}{100} \right)$$

  - Status states for every earnings line: Accruing (Work in Progress), Invoiced to Client, Approved for Payout, Disbursed.

3.4 Task Verification & Audit Logging System

- **Checklist Execution Tracker:**
  - Every task supports multi-item granular checklists.
  - When a user checks or unchecks an item, the system triggers a background audit capture.
- **Audit Trail Immutability:**
  - Log records are append-only. No user—including the Super Admin—can alter or purge audit logs from within the application interface.
  - Captured Event Data:
    - Log ID: UUIDv4
    - Task ID & Project ID
    - Actor ID (User performing the action)
    - Action Type (CREATED, MODIFIED, STATUS_CHANGED, ITEM_CHECKED, ITEM_UNCHECKED, DELETED)
    - Previous State vs. New State (JSON delta)
    - Client IP Address and User-Agent String
    - Timestamp (UTC, ISO-8601 down to milliseconds)
- **User-Facing Activity Stream:**
  - An expandable activity panel embedded in the task detail view displaying real-time chronological actions: *"Jane Doe marked 'Implement JWT Middleware' as completed on 2026-10-01 14:15 UTC"*.

4. Technical Architecture & Database Schema

4.1 Recommended Tech Stack

- **Frontend:** Next.js / React with Tailwind CSS
- **Backend:** Node.js (NestJS)
- **Database:** PostgreSQL (Relational schema with JSONB support for audit deltas and Row-Level Security for multi-role scoping).
- **Caching & Real-Time:** Redis (Pub/Sub for real-time task checklist sync) and WebSockets.

5. Non-Functional Requirements & Security Specifications

5.1 Security & Access Partitioning

- **Database Row-Level Security (RLS):** Apply PostgreSQL RLS policies to tasks, projects, and payout_distributions so that users possessing the FREELANCER role cannot retrieve rows belonging to unassigned projects.
- **Audit Trail Integrity:** No database grants for UPDATE or DELETE on task_audit_logs are allocated to the application server user role. Entries are append-only.
- **Authentication:** Multi-factor authentication (MFA) required for Super Admin accounts; salted Argon2id/bcrypt password hashing; token revocation lists for JWT sessions.

5.2 Performance & Reliability

- **Throughput:** System must handle concurrent checklist completion updates with sub-250ms round-trip latency.
- **Event Dispatch:** Audit log captures must execute asynchronously or within lightweight background database triggers to avoid blocking the main task state update pipeline.
- **Backup & Disaster Recovery:** Point-in-time database recovery (PITR) with automated daily offsite snapshots

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